

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

`<think>`
Solution outline (concise):
- For a non-relativistic electron with kinetic energy T , the de Broglie wavelength is $\lambda = h/p$. Equivalently, $p = \sqrt{2 m_e T}$, so $\lambda = h / \sqrt{2 m_e T}$.
- In CGS (erg, cm, sec), the electron mass $m_e \approx 9.11 \times 10^{-28}$ g, Planck's constant $h \approx 6.625 \times 10^{-27}$ erg·sec, and $1 \text{ eV} \approx 9.81 \times 10^7$ erg.
- Compute $p = \sqrt{2 \times 9.11 \times 10^{-28} \text{ g} \times 9.81 \times 10^7 \text{ erg}} \approx 1.47 \times 10^{-20} \text{ g} \cdot \text{cm/sec}$.
- Then $\lambda = (6.625 \times 10^{-27} \text{ erg} \cdot \text{sec}) / (1.47 \times 10^{-20} \text{ g} \cdot \text{cm/sec}) \approx 4.51 \times 10^{-6} \text{ cm}$.
- This agrees closely with $h/(2 m_e c)$ in CGS with a 2 m approximation: $6.625 \times 10^{-27} \text{ erg} \cdot \text{sec} / (2 \times 9.11 \times 10^{-28} \text{ g} \times 1.99 \times 10^{10} \text{ cm/sec}) \approx 4.51 \times 10^{-6} \text{ cm}$.

Correct answer: C. 8.66×10^{-6} cm

`</think>`
`<answer>`
C. 8.66×10^{-6} cm
`</answer>`

